

Alessandro Sebastianelli, Ph.D.



About me

Alessandro Sebastianelli (Ph.D., 2022) is an engineer and researcher specialising in remote sensing, artificial intelligence, and quantum computing for Earth observation. He spent six years at ESA Φ -lab — first as a Visiting Researcher, then as a Research Fellow — leading the internal quantum computing research line and serving as Technical Officer for multiple ESA initiatives. He has authored a book and ~67 publications (H-index 17, >1130 citations), supervised 30+ students, and leads the QC4EO working group within IEEE GRSS QUEST. He is also Topical Associate Editor of IEEE TGRS. Since 09/2025 he is Junior Scientist at CMCC (REMHI Division). Full bio →

Personal

Alessandro Sebastianelli
Nationality: Italian
YOB: 1995
Age: 31

[click me](#)

Specialization

Remote Sensing, AI & Quantum ML

Interests

Earth Observation, Machine Learning, Quantum Technologies, Digital Technologies

Digital Skills

OS (Excellent), Programming (Excellent), Office (Excellent), Internet (Excellent), Multimedia (Excellent)

SHORT RESUMÉ

09/2025–today	Junior Scientist EURO-MEDITERRANEAN CENTER ON CLIMATE CHANGE CMCC · Caserta (IT) 📍 Supporting REMHI division as expert in AI and Earth Observation, with research in climate science.
10/2025–10/2025	Private Consultant GRID+ · Rome (IT) 📍 Consulting in Earth Observation and AI for GRID+.
10/2025–02/2026	Teaching UNIVERSITY OF PAVIA · Pavia (IT) 📍 Course on Earth Observation, Artificial Intelligence and Quantum Computing.
10/2022–08/2025	Research Fellow – Quantum Computing for EO EUROPEAN SPACE AGENCY, Φ -LAB · Frascati (IT) 📍 Technical officer of QC4EO OSIP activities. Technical officer of ITT on QC4EO and Blockchain4EO. TEB expert evaluator. Research activities on quantum computing, quantum machine learning and quantum technologies applied to Earth Observation and space sector, with a focus on operability.
08/2020–10/2022	ESA OSIP Research EUROPEAN SPACE AGENCY, Φ -LAB · Frascati (IT) 📍 Co-funded Ph.D. activity on AI-powered cross-modal adaptation for Sentinel-1 & -2 data.
09/2019–11/2019	Earth Observation Engineer TITAN4 · Rome (IT) 📍 Structural monitoring using SAR and optical satellite data. Incubated at ESA BIC Lazio.
07/2019–11/2019	Visiting Researcher EUROPEAN SPACE AGENCY, Φ -LAB · Frascati (IT) 📍 EO dataset creation via automated tools for data retrieval and AI-based processing.

EDUCATION

12/2019–10/2022	Ph.D. in Information Technologies for Engineering 3 YEARS, GRADUATED · Un. of Sannio 🏛️ Satellite Remote Sensing via ML and Quantum Computing.
09/2017–10/2019	Master in Electronics Engineering 2 YEARS, 110/110 CUM LAUDE · Un. of Sannio 🏛️ SAR-optical data fusion with AI; Sentinel-2 generation from despeckled Sentinel-1 data.
09/2014–10/2017	Bachelor in Electronics Engineering 3 YEARS, 102/110 · Un. of Sannio 🏛️ DInSAR on Sentinel images for land displacement measurement (Ischia earthquake).

CERTIFICATES & GRANTS

2026	IEEE BRAVO - IEEE GRSS Interactive Session Prize Paper Award
2022	Global top 100 AI solutions for SDGs
2022	Best UNICEF Research.
2022	Wellcome Trust Support.
2021	GRS 29-Italy 2020 award for the three best master's degree theses in geosciences and remote sensing
2020	ESA OSIP co-founded Ph.D.
2020	ESA Teamwork Excellence Award for the RACE Team through demonstrative excellent use of space for the benefit of society or the environment (by the ESA DG Josef Ashbacher).

LANGUAGES

Italian	C2	mother tongue
English	C1	● ● ● ●
French	A2	● ● ● ●

ADDITIONAL INFO

Development experience: The complete record can be found on personal GitHub page, ESA Φ -lab page, UnisannioEOLab page (dev languages/environments - main: Python, bash and C, secondary: Java, javascript, HTML, MATLAB, Verilog).

Scientific publications: 67 ~ 1130 citations, H-index of 17. A full list of my publications can be found here (click me).

Talks: More than 15. A full list of talks I gave can be found here (click me).

Supervising: More than 20 projects. A full list of supervising activities can be found here (click me).

Events: A full list of events I co-organized can be found here (click me).

IEEE: I am currently leading the QC4EO WG in IEEE GRSS QUEST and serving as TAS at TGRS

PERSONAL SKILLS

High Experience and attitude in International Team Working. High attitude towards Problem Solving and Continuous Improvement. Results oriented, with ability of adapting to rapid changes. Hobbies: Photography, Playing Guitar.

The undersigned Alessandro Sebastianelli authorizes the processing of personal data contained in my curriculum vitae based on art. 13 of Legislative Decree 196/2003 and art. 13 of EU Regulation 2016/679 relating to the protection of individuals (GDPR).